

Subject Name: Microbiology Laboratory

Subject No: BT29205

Prerequisite: Microbiology (BT20205)

Objective

Microbiology is the study of microorganisms (bacteria, virus, fungi and protists) and it focuses on their structure, function, classification and also in terms of controlling their growth and activities that can be explored for various studies. In a range of controlled settings, microorganisms can be exploited for their beneficial use in the areas as diverse as healthcare, food production, agriculture, etc. Microbiology laboratory aims at providing the basic and fundamental knowledge on the techniques used to achieve the above-mentioned goals. Not only it enables the students to use different laboratory instruments and tools effectively, but also ensures that the students should learn safe handling of the potentially pathogenic organisms.

LTP - 0-0-3

Credit - 2

Content:

Each experiment of 3 hr duration:

1. Introduction to basic microbiological tools, terms and practices:
 - Identification, preparation and uses of different growth media used in microbiology laboratory
2. Isolation of pure culture in slant technique:
 - Inoculating microbial culture on a solid medium
 - Identification of colonies of pure culture
3. Isolation of pure culture by streak plate technique:
 - Introduction to streaking technique
 - Identification and isolation of pure colonies
4. Dilution plating for viable count:
 - Introduction to spreading technique
 - Calculating approximate number of microbial cells present in a microbial culture
5. Simple staining of bacteria:
 - Identification of different parts and proper handling of a light microscope
 - Preparation of bacterial smear
 - Use of simple stain, basic principle and technique of staining bacteria
6. Gram staining of bacteria:
 - Differentiating bacteria on the basis of cell wall chemistry
 - Basic principle and technique of the differential staining procedure using Crystal violet and Safranin
7. Size measurement of a microbial cell:
 - Introduction to micrometry and micrometers

- Calibration of ocular micrometer using a stage micrometer
 - Determining the size of a microorganism using micrometry
8. Determination of cell number:
- Introduction to Hemocytometer
 - Cell counting on a hemocytometer under a microscope
 - Calculating approximate number of microbial cells present in a microbial culture
9. Staining the bacterial endospore:
- Principle and technique of staining bacterial spores using Malachite green
 - Counter staining vegetative bacterial cells with Safranin
10. Determination of antibiotic sensitivity of *E. coli*:
- Brief description of different antibiotics and their mode of action on different bacteria
 - Use of MULTIDISC in determining sensitivity of the bacteria *E. coli* against different antibiotics
11. Determination of minimum inhibitory concentration (MIC) of antibiotics against *E. coli*:
- Determining the MICs of specific antibiotics required to inhibit the growth of *E. coli*
12. Enumeration of heterotrophic bacteria from drinking water:
- Collection of water samples from different sources
 - Determining the type and nature of bacteria present in them for a comparative and qualitative study

Text Book:

1. Microbiology: A Laboratory Manual by James G. Cappuccino and Natalie Sherman; Benjamin Cummings, 9th Edition, illustrated, 2011; ISBN 0321673875, 9780321673879